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Perceived security determinants in e-commerce among Turkish university students

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Perceived security is defined as the level of security that users feel while they are shopping on e-commerce sites. The aims of this study were to determine items that positively influence this feeling of security by users during shopping, and to develop guidelines for perceived security in e-commerce. An experiment allowed users with different security assurances to shop on simulated e-commerce sites. The participants were divided into three groups, shopping for cheap, mid-range, and expensive products, respectively. Following the shopping environment, the virtual shopping security questionnaire (VSSQ), consisting of fourteen perceived security items, was presented to the users. The VSSQ was presented to the participants to validate these perceived security items. The VSSQ had a Cronbach's alpha internal reliability value of 0.70. With the exception of two items, there were no significant differences in item ratings between the groups of different shopping item values. A factor analysis procedure determined two main factors concerning perceived security in e-commerce. The perceived operational factor includes: the site's blocking of unauthorized access; emphasis on login name and password authentication; funding and budget spent on security; monitoring of user compliance with security procedures; integration of state-of-the-art systems; distribution of security items within the site; website's encryption strategy; and consolidation with network security vendors. The perceived policy-related factor includes: the website's emphasis on network security; top management commitment; effort to make users aware of security procedures; the website's keeping up-to-date with product standards; the website's emphasis on security in file transfers; and issues concerning the web browser.

1. Objective and significance

With internet crime increasing at geometric rates in the last several years, security has become one of the most important issues in business to consumer (B2C) e-commerce. Although extensive research has been conducted in computer science terminology on technology development for security in e-commerce, research involving the user's perspective regarding e-commerce security is minimal. This research aims at exploring the issues that users deem as important in giving them a secure feeling as part of their online shopping experience. The security issues that carry significant importance from the users' perspective, and the interrelations between these issues are identified as part of determining what can be done by online businesses to improve the overall security feeling of users while they shop. Recent research indicates that a very high number of users in Turkey (more than 50%) are very

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hesitant in giving their credit card numbers online because of the fear that the information can be stolen by third parties, and the lack of a feeling of security is the primary reason for their not shopping online at all (Lightner *et al.* 2002). Therefore, this study intends to quantitatively determine the issues that the electronic shopper can be made aware of, and that would result in an improved feeling of security. The term 'perceived security' is therefore created to indicate how secure users feel while shopping online. It should be noted that users' perception of security can be very different from the real security level on the e-commerce site. A secure site may give no indication of security precautions being taken, and a highly insecure site can give misleading impressions to users that they have very good security. While laws in the US require e-commerce companies to have substantial security technologies in place in order to conduct business, there are no unified laws regulating security items in international e-commerce. The Turkish population was chosen to explore a different e-commerce environment than the relatively mature environment in the US. Additionally, the findings can be compared to perceived security determinants in the US in a future, cross-cultural study. A group of mostly male engineering undergraduate students from a leading university in Turkey have been chosen as the participant group. The engineering student group was chosen because the study by Lightner *et al.* (2002) determined that college students are a highly representative group of typical online shoppers in Turkey, mainly due to their familiarity with computers and e-commerce. A severe difficulty in recruiting female participants at the time of the study resulted in a fairly large number of male participants (86% of the overall sample). Therefore, the results of the experiment can arguably be applicable to a largely male engineering student population.

An identifying dimension of perceived security is determined by the researchers to be the prices of products that are being purchased online. The relationship between the prices of the items shopped for and the perceived importance of the security factors is explored in the study, since a logical hypothesis could be that the security issue gains more importance as the items shopped for get more expensive. This study aims to determine the factors that are of significant importance to Turkish, college-age shoppers in feeling secure online, whether there are any categorizations or interrelationships between these factors, and whether there is a relationship between the expense of the goods shopped for and the level of emphasis (importance) given to security by the electronic shopper. Previous research indicates that there is next to no security precautions on e-commerce sites emphasized to shoppers. This study also aims to develop e-commerce page guidelines for maintaining a high perception of security by Turkish, college-age shoppers, thereby possibly improving the sites' business.

It should be noted that brand name recognition and company reputation are significant elements of trustworthiness in the online shopping environment. For example, there is almost no doubt that Amazon benefits a great deal from its good reputation as a safe online store. Additionally, recent practical advances in technology have allowed several new options and certifications to boost security in online shopping environments, including security seals offered by VeriSign, alternative shipping and payment options, and opt-out options for private information.

It should be noted that the concept of perceived security is different from the concept of trustworthiness. Trustworthiness of a site is how much the users trust the site based on items including brand name recognition, on time product delivery, fraud protection and more (Fogg *et al.* 2001, Grabner-Krauter and Kaluscha 2003). In most cases, a site can gain trustworthiness based on its long-term performance and through word-of-mouth by its users. Perceived security, on the other hand, deals with the security feeling of the shoppers while on the site only. It is not a concept that is gained by the shopping site over a long period of time, but rather depends on the security information that is presented to the shopper or user while they are on the site. Therefore, the researchers have concluded that trustworthiness and perceived security are two separate concepts, although a site providing high perceived security for its shoppers may over time also boost its trustworthiness. The trustworthiness items are out of the scope of this research, mainly because this research is focused on items that are presented to the user by the online company, on the interfaces (screens) the user is shopping at. In this context, for example, items that are presented additionally as part of post-shopping activities (e.g. shipping/payment options) and offered by third parties (e.g. security certifications) are not studied in this research.

2. Literature review

Although previous literature deals closely with security and fraud prevention, how customers in e-shopping environments can be provided with a perception of a high level of security has not been explored in e-commerce and human–computer interaction literature. While the concept of perceived security is not clearly defined in the literature, there is a body of work dealing with user behavior stemming from security. Miyazaki and Fernandez (2000) identified two categories regarding security in e-commerce affecting user behavior: recording and use of customer information, and third party intrusion. They indicate that the surge in e-commerce has resulted in a dramatic increase in the information regarding the customers collected by vendors, which can be with or without the consent and knowledge of the customer. And while few customers knew about their information being collected, recorded and

possibly used for purposes that may be undesired during the growth of populist e-shopping (such as selling their contact information to other vendors) or illegal (such as being used for identity theft), consumers are becoming more and more conscious of how their information is being used. Moreover, they are becoming increasingly hesitant about giving out sensitive information on the web. In this context, a major ethical issue that has been identified is the collection and management of consumer information even before the inception of e-commerce in the mid-to-late 1990s (Jones 1991, Foxman and Kilcoyne 1993, Bloom *et al.* 1994, Chonko 1995. Similarly, Rohm and Milne (1998) indicated that as of 1998, a majority of internet users, both those who make purchases and those who do not, had concerns regarding information privacy.

In addition to privacy, the concepts of information theft and misuse (mostly involving sensitive personal information such as credit card numbers) have been identified as major security threats in e-commerce by government and consumer organizations (Brinkley 1998, National Consumers League 1999), as well as the popular press. Furthermore, it is believed by some researchers that the tremendous media coverage of security threats is a major factor that to some extent discourages full consumer adoption of e-commerce (Judge 1998).

A comprehensive list of barriers facing e-commerce adoption has been summarized by Furnell and Karweni (1999). They indicated that because of the global nature of the internet, trust is more important in e-commerce than traditional commerce, mainly because the party being dealt with may be unknown; it is not possible to have full control of the data during the transfer (a fact that results in many users tracking the products online but then giving the actual order via telephone or fax); and the possibility of the other party being at a different, unknown location, and thereby being subject to different laws and legislation. Ratnasingham (1998) indicates that it is therefore necessary to facilitate a 'complete trustworthy relationship' among the trading partners. The requirements to achieve this relationship include a third party that is deemed trustworthy by both sides to mediate if one of the parties is not known directly; data being secured at all stages; and common rules or at least a known and acceptable legal environment having been established. In their study, Furnell and Karweni (1999) had the goal of determining people's attitudes, awareness and expectations in terms of e-commerce security, and they identified secure communication, authentication of the customer by the vendor, authentication of the vendor by the customer, precautions against fraudulent actions, and non-repudiation (the inability for both sides to argue that it was not them who participated in the transactions) as the most essential items from the customers' perspective to feel secure while shopping. From the perspective of information sharing on the internet, a study conducted by Ackerman *et al.* (1999) indicated that while a majority of internet users did not mind sharing relatively public information such as favorite TV show (82%), age (69%), full name (54%) and postal address (44%), the users were significantly less willing to share information that is more sensitive in nature, including social security number (1%), credit card number (3%), and phone number (11%). It should be noted that, while 76% in the study were willing to share their e-mail address on the internet, this number is likely to have been strongly reduced because of the recent dramatic increase in spam messages. Another study, conducted by Udo (2001), indicated that the major customer concerns regarding shopping online included, from highest level to the lowest, privacy concern, security and threats concern, children protection concern, e-mail safety concern, censorship concern, and impersonation and forged identity concern. Similar to the previous study, it should be noted that in the years since the study by Udo, identity theft on the internet has increased dramatically, and it is therefore likely that concern regarding forged identity has significantly increased since.

To address these and many other issues, the e-commerce vendors responded. Nowadays, secure transactions using encryption for all e-commerce activities are mandated by US law (Consumer Reports Online 1998, Federal Trade Commission 1998), and a very large percentage of online retailers are offering consumer guarantees against credit card fraud (National Consumer League 1999). Other security precautions taken by e-commerce vendors include offering alternative payment options to the customer, zero liability credit cards, credit card escrow services, security certifications such as eTrust, and additional credit card security checks by the e-commerce vendors. It should be noted that these items are put in place by the vendors not with the direct intent of improving perceived security, but rather to increase customer acquisition and retention, among others. This research intends to determine the factors that users can be made aware of to improve perceived security, regardless of whether these items were placed on the site with the specific intent of improving perceived security or not. In this same context, Miyazaki and Fernandez (2000) concluded that the examination of online disclosure practices should be an ongoing research effort with respect to how such practices may affect consumer perceptions, and instead of examining only perceived risk toward general online shopping, specific assessments of risk regarding online privacy, retailer fraud and security of online transaction systems could be helpful for understanding the aspects of overall perceived online security. Additionally, Miyazaki and Fernandez (2000) indicated that future research should also cover how government security mandates may affect the overall security perceptions of online shoppers.

When the previous literature was surveyed, it became oviously apparent that privacy protection was one of the

overwhelming issues affecting perceived security. In this context, Oppliger (2000) indicated that because activities on the World Wide Web are very easily observable, e-shoppers are becoming discerning regarding the exchange of their personal information online, and also prefer to stay anonymous unless they absolutely have to reveal their identity. According to Oppliger (2000), for this same purpose, some electronic payment systems allow secure transactions over the internet while preserving the untraceability of cash. However, these third-party payment methods have recently invoked new fears concerning the possibility of illegal transactions being conducted over the internet without being traced. It is therefore doubtful, and there has been no evidence so far indicating, that the concept of e-cash, which also never really took off in popularity as a payment and money transfer method over the internet, can be a significant factor boosting perceived security in e-commerce transactions. However, fundamental anonymous communication properties should be mentioned within the context of security, which include sender anonymity, receiver anonymity, and the inability to link the sender and the receiver (Pfitzmann and Waidner 1987). It should be noted that, while in the majority of internet transactions, information concerning the three items are available to selected parties (mostly payment providers), there is no evidence in the literature indicating that being totally anonymous while doing e-commerce transactions is a factor that improves perceived security in e-commerce, probably due to the fact that the concept of being anonymous is almost incompatible with e-shopping. Although there are third party services providing 'anonymity' by hiding user information from vendors, the user information is still kept within the third party providing anonymity. Since for all payment transactions, user information needs to be available to one of the parties, it is a logical conclusion that 100% anonymity cannot be provided to e-shoppers and is therefore not a direct factor of perceived security.

Looking at Web security in general, Rubin and Geer (1998) indicate that security on the Web was rather an 'afterthought' than an integral issue, and vendors engaged in retrofitting security must contend with the Web environment's peculiarities including communication between parties that don't know each other and privacy and anonymity. Also, a survey by Dowland *et al.* (1999) identified two items affecting perceived security on the internet: public awareness of, and attitudes towards, computer crime and abuse; and the influence the media has over individual views and perceptions of computer crime and abuse. It should be noted that this particular study indicates the media attention on security and security failures as a major factor negatively influencing perceived security.

In summary, the literature survey hinted at an indirect gain through perceived security, and its importance in e-commerce, and while the security items in e-commerce from the users' perspective have been identified by some studies, the exact elements that affect the perceived security of e-commerce shoppers have not been defined. Indeed, instructions or guidelines for improving perceived security in e-commerce have never been developed.

3. Experiment and survey design

For the reasons explained in the last section of the literature review, the current study aims at filling this gap in the literature, namely, developing a comprehensive list of perceived security items as well as guidelines to provide perceived security in e-commerce. For this purpose, a survey was developed to identify the elements and categories of perceived security in e-commerce, exactly. Before the survey was administered to each participant, an experiment called the 'virtual shopping experience' allowed them to become familiar with possible security precautions that can be presented by e-commerce vendors. This experiment, and its procedure, are explained in the next four sections. The virtual shopping experience was designed requiring users to go through different, simulated shopping sites, and a survey was designed to explicitly determine the different elements and categories of perceived security. The primary reason for designing the experimental environment was to simulate a shopping experience for three different groups of shoppers and to familiarize them with potential security precautions that can be taken by e-commerce sites and presented to the shoppers. One group was provided with a shopping environment where they shopped for cheap products, one group for middle-price-range products, and one group for expensive products. Immediately following the shopping experience environment, participants from each group were provided with a survey concerning perceived security items. The goal for this procedure was to detect the differences in perceived security items between the cases when users were shopping for cheap, mid-priced, and expensive items. In the next sections, the experimental design and survey are discussed.

3.1 *The virtual shopping experience*

The virtual shopping experience (VSE) was designed by the researchers to simulate an online shopping environment for the participants of this study. It was designed and implemented in Turkish, the native language of Turkey. The goal of the VSE was to introduce an online shopping environment to the participants where different websites had different security measures presented to the online shopper, and through these presentations, to familiarize the users with common types of security measures in e-commerce. As an additional measure, the total performance time for each participant to complete the entire

shopping procedure was recorded to determine any correlations available between performance time and security scores obtained from the questionnaire (explained in the next section).

Two main issues were of interest in the experiment. The first issue was shopping for products of different values since one of the main goals of this research was to determine the relationship between the security expectations of e-shoppers from e-commerce sites and the prices of the goods they shop for. The second issue was to determine the elements and categories of perceived security in e-commerce. To empirically determine the first issue, three groups were identified in the experiment. Each group were presented with a shopping experience within a certain price range per shopped product. Participants of Group 1 shopped in a simulated environment for cheap products ($5–10 range), participants of Group 2 shopped in a simulated environment for products of mid-range price ($30–100), and participants of Group 3 shopped in a simulated environment for expensive products ($300–1000). The typical products that were shopped for in Group 1 included CDs and mainstream books. The typical products shopped for in Group 2 included small electronic devices such as calculators, low-end personal digital assistants (PDAs), clocks and watches. The typical products shopped for in Group 3 included home appliances such as refrigerators, large-screen (27-inch and up) TVs and high-end DVD players (DVD players are significantly more expensive in Turkey than in the US). One important issue of caution is the relationship between the price of items and the income of the participants as an indication of affordability of the goods. To keep the affordability at a reasonably stable range, the variance of the income among the participants was kept at a low level, with an average of about $1000 dollars per month, which is a representative number for the average monthly household income in big cities in Turkey. The products chosen for Groups 1 and 2 are relatively representative types of consumer products (up to $100) targeting the younger college population in Turkey. The products in Group 3 (up to $1000) are less representative since they are highly expensive products especially compared to the average household income of the target population. However, they are still included, because, while students may shop for them online in very rare occasions, determining the relationship between product pricing and security requirements from shoppers is one of the goals of this study. Additionally, though rarely, students may still shop for these products online for themselves or their families, especially taking advantage of the frequent online discount offers for such products. The particular scope of population (college students) and products (CDs, books, electronic devices, household appliances) were chosen because they are representative of a common shopping population and common products being sold, respectively, in the Turkish online environment.

Each group contained a total of six different shopping sites, from each of which the software instructed the participants to purchase one item of their liking, resulting in a total of six shopping actions. As indicated previously, the main goal of the VSE was to familiarize the participants with possible different security items the e-commerce companies present to them. Six different simulated shopping sites created by the researchers presented different layers of security to the participants to indicate to them what the site is doing to ensure their shopping security. This was done in order to fulfill the goal of the second issue identified above. Each site included two different security precautions that were presented to the user. The first site included a message that indicated what encryption was used on the site for security. The second site included a message that read 'You are now entering a secure site. All unauthorized access is blocked to this site.' The third site included warnings in the form that all secure areas on the site had a green background while all insecure areas included a red background. The fourth site included a brief paragraph on the first page regarding the security policy and procedures on the site. The fifth site included a paragraph in the payment section ensuring the user that the credit card information is transferred using highly secure encryption. Finally, the sixth site included a message presented to the user concerning the specifications of the web browser they are using, such as 'You are using a 128-bit Netscape browser.' These security items were adopted from some of the items in the questionnaire in order to familiarize the participants with them before they filled out the questionnaire. The order with which these security items were presented to each participant was randomized along with the order of presentation for each website containing these items.

3.2 *Participants*

The experimental design was a 3-group, between-subject design, with ten participants in each group. Thirty college students from the School of Engineering at the Technical University of Istanbul were recruited for the study, with a mean age of 21.0 and standard deviation of 3.11. Twenty-six (87%) of the participants were male, and four participants (13%) were female. Twenty-five of the participants were from the Department of Industrial Engineering, two participants were from the Department of Mechanical Engineering, one participant was from the Department of Electrical Engineering, one participant was from the Department of Geological Engineering, and one participant was from the Department of Management Engineering. The average family income for the thirty participants was 1.46 billion Turkish Lira which would translate to just about $1000 per month based on the currency rate at the time of the study (summer 2002). All of the participants indicated that they have shopped online

before. Nineteen participants indicated they shop online less than once every two weeks. Six participants indicated they shop online once a week to once every two weeks, and five participants indicated they shop online at least once a week. All of the demographic data presented above were collected from a pre-experiment demographic questionnaire, and using the data collected from this demographic data, the participants were balanced in terms of gender and shopping frequency when they were assigned to each of the three (cheap, mid-range, expensive goods) groups, in order to avoid any effects that may result from significant age or online shopping frequency differences among the groups.

It should be noted that the participant body contained a relatively small percentage of females. This was because of difficulties in recruiting female participants at the time of the study, due to reasons that could not be determined by the researchers. While the four female participants were distributed into the three groups relatively evenly (two to Group 1 and one to Groups 2 and 3), the overall number of the female participants was arguably low due to this practical difficulty (this issue is addressed in section 6).

3.3 *Procedure*

The experiment was conducted in an experimental room, in a setup adequate for fast display of the pages. The experiment consisted of three main stages: in the first stage, after the participants were informed regarding the experimental conditions and a consent form was signed, a pre-experiment questionnaire was presented to each participant online to collect demographic information. This information was checked immediately by the experimenter, and participants who had never shopped online before were eliminated from the experiment. Next, each participant was randomly assigned to one of the three groups, and the experimenter shortly instructed the participant to shop for six different products of their liking and inform them when they were done. Next the experimenter left the room for the participant to complete the virtual shopping experience. Once the participants completed the virtual shopping experience, the experimenter presented the virtual shopping security questionnaire (VSSQ) to the participant on the computer, gave a short explanation of the questionnaire, and waited in the same room as the participant, but did not interfere with the participant's filling out of the questionnaire. Once the participant completed the questionnaire, they were thanked and ushered out of the room.

3.4 *The virtual shopping security questionnaire (VSSQ)*

The virtual shopping security questionnaire (VSSQ) consisted of fourteen items (questions 1 – 14) measuring the importance of the security precautions taken by e-commerce sites from the e-shopper's perspective. Two additional items (questions 15 and 16) were inserted in the questionnaire for internal reliability purposes. Cronbach's alpha coefficient was used to measure the internal reliability of the VSSQ. The Cronbach's alpha coefficient for the questionnaire was 0.70. Although this value is relatively low, a number of researchers accept 0.70 as the acceptability threshold (Cronbach 1990) for internal reliability of a survey. Therefore, it was concluded that the questionnaire had acceptable internal reliability.

The VSSQ was designed and implemented in Turkish. It was designed by an expert in English linguistics who received both his Master's and PhD degrees in the area of linguistics and who is a native Turkish speaker. An additional native Turkish speaker who has extensive engineering training in English language overlooked the design procedure. This process ensured that the language and concepts of the VSSQ were consistent with the understanding of the Turkish sample group. Also, the participant group had good familiarity with online shopping, and prior to answering the questionnaire, the participants were strictly instructed to ask the experimenter any questions about the questionnaire items that they didn't understand. The researchers believe that with these precautions, the possibility of participants' filling out some questions without exactly knowing the content of those questions was eliminated. Additionally, the language expert who designed the VSSQ translated the survey into English during the design process to ensure that the items of the survey had English counterparts that were applicable to English-speaking populations. This allowed the researchers to produce the English version of the VSSQ at the same time as the Turkish version. The items in the VSSQ were based on the most common issues in the literature that related to the feeling of security among electronic shoppers to ensure construct validity.

The participants were asked to rate the importance of each of the individual items of security in on-line shopping which they regarded with utmost importance. To best detect the differences in participants' ratings among the different perceived security items, a 7-point Likert scale was used, with a score of 1 indicating the item is 'not important at all', a score of 4 indicating the item is 'neither important nor unimportant', and a score of 7 indicating the item is 'extremely important'. The fourteen questionnaire items, and discussions concerning each questionnaire item's relevance to the concept of perceived security, follow:

3.4.1 The website's blocking of unauthorized access into its quarters by hackers. Unauthorized access can be defined as third party intrusion and access to sensitive information. Malicious third party intrusion involves several categories. The first category is active intrusion with a programmer (a

hacker) sitting on the other end of the connection and breaking into the company website and stealing information, either dynamically while an electronic transaction is in progress, or stealing static information from company databases. Another category is taking the same actions of information stealing, but doing it by sending malicious codes in the forms of worms and Trojan horses to get sensitive information from company databases. Companies' blocking of those intrusions mostly rely on firewall and intrusion detection technologies. But in addition to implementing such applications to monitor and eliminate such malicious intrusion, Miyazaki and Fernandez (2000) indicate that these activities by companies to keep hackers away from the website headquarters improve the users' trust of the company's shopping site. It is therefore determined that making users aware of these activities is a strong item in improving perceived security and this particular item has been included in the survey.

3.4.2 The website's emphasis on network security. A more general aspect in nature, this particular item has been generated based on the common belief that a network's security emphasis directly affects consumer-perceived security. The inclusion of this particular question aims at determining the validity of this belief.

3.4.3 The website's top management commitment in security-related issues. As part of growing consumer awareness in security-related issues concerning e-commerce, companies' engagement with security at a top level has been indicated to be a major item in consumer's preferences for shopping sites, as indicated by Dowland *et al*. (1999), among others. Therefore, previous literature presents a top-level management involvement with security as an item that the consumers would like to be aware of in order to determine the overall security level of the sites they are shopping at. To validate these indications by previous literature, this item has been added to the virtual shopping security questionnaire.

3.4.4 The website's effort to make its users aware of the security procedures it utilizes. This particular item has been made part of the customer questionnaire, particularly because the fundament of this research article heavily relies on the hypothesis that the site's making the users aware of the security measures it is taking significantly and positively affects the overall perceived security level. Previously, it has been discussed that users' perception of security does not only depend on security precautions being provided by the website, but also on the website's making the users aware of these precautions. Therefore, the question has been included to determine if, and to what level, user security awareness results in the user's perception of a secure e-shopping environment.

3.4.5 The website's emphasis on login name and password request. Login name and password-based authentications are ubiquitous in accessing all types of personal information on the web. However, the precise occasions where passwords and login names are required vary greatly between websites. While all of the shopping sites require the user to register to create a login name and password to track their order status and for all of the account information modifications, for example, some sites ask for a password again if the user has previously logged in, is intending to change the user information, but has been doing other actions on the site between first login and the request to change user information. In this example, this extra password re-verification can be interpreted as a security precaution against the user having logged in, not having closed the window, and another user possibly continuing on the user's open browser window, thereby possibly gaining access to the user account. This particular item is inserted into the questionnaire to determine the importance of login name and password authentication systems in perceived security.

3.4.6 The website's funding and budget spent on security. Similar to items 3.4.2 and 3.4.3, the budget and funding spent on security can be thought of as direct, positive items that improve perceived security. The insertion of these items into the questionnaire aims at determining whether, and to what degree, making the users aware of the website's monetary spending concerning security would improve the overall secure feeling of the users while shopping online.

3.4.7 The website's monitoring whether the users are complying with the security procedures. This particular item is parallel to item 3.4.1 in its intent to block third party intrusion, since malicious hackers can pose as customers to cause security breaches. On the other hand, constantly being monitored by the site may also be perceived as an invasion of privacy by the user. Therefore, this item was included in the questionnaire to determine whether monitoring the user is an item that boosts or reduces the security feeling of the e-shopper.

3.4.8 The website's distribution of security items within its body. This item relates to how the security items are distributed and used within the site. It is believed that the security items being consistently provided within the different pages of the site is an issue that would positively affect perceived security, and this item has been added for that particular reason.

3.4.9 The website's integration of state-of-the-art security systems. Again, this item can be interpreted as very close to items 3.4.2, 3.4.3 and 3.4.6. It has been indicated in this and previous chapters that management involvement at a higher

level is one of the items likely to improve perceived security. In addition to this strong indication of management involvement in security being essential, many researchers such as Mott (2000) indicate the importance of the continuance of new technology and standards development concerning security in solving consumer trust-related issues. Therefore, this question has been included in the questionnaire with the intention of determining how making the user aware of the inclusion of new and reliable security systems affects the users' perceived security.

3.4.10 The website's keeping up-to-date with product standards. This item has been included in the questionnaire as an extension of item 3.4.9 to more emphasize the issue of being 'state-of-the-art' in consumer security. This particular item has been included to determine, aside from having the latest security systems, to what extent a constant update of the security systems and product standards would affect the perceived security of the users. This updated state of security items can be presented to users with messages including the version details of the security software and hardware being used on the website, also indicating they are the latest versions.

3.4.11 The website's emphasis on security in file transfers over the internet File transfers are highly common in purchasing electronic products as downloads in e-commerce. Secure file transfer protocols (FTP) are commonly used in internet transactions, and therefore this item is intended to discover how much of an emphasis users give on security in file transfers to feel secure during their transactions.

3.4.12 The website's encryption strategy. While many standards are available in encryption, it is very difficult to name a hacker-proof encryption type (Mott 2000). Encryption technologies need to be constantly updated in order to provide a permanently secure environment. The encryption type and strategy of a website in e-commerce can be presented to the users as messages during transactions. Therefore, this item is intended to determine the emphasis of the users on encryption in general when they are using the website for shopping purposes.

3.4.13 Issues concerning my web browser. Browsers' acceptance of site encryption plays a major role in secure transactions. Today, most e-commerce companies do not allow users to conduct transactions with web browsers not supporting less than 128-bit encryption. Therefore, it is the intention of this item to determine the importance of browser-related security messages in perceived security.

3.4.14 The website's consolidation with network security vendors. This particular item directly relates to items 3.4.3 and 3.4.6 in particular, and is included for the same reason as those items, i.e. to understand whether the website owner's consolidation with security vendors affect the e-shoppers' perceived security.

4. Data analysis

4.1 *Analysis of the relationship between the prices of goods and the user emphasis on security items*

One of the primary goals of this research was to determine whether the emphasis on perceived security items changed when the value of shopped items increased. To determine whether increasing the prices of the shopped products affected the subjects' emphasis on and/or requirements of perceived security items, analysis of variance (ANOVA) procedures were conducted for each of the fourteen security items, presented by fourteen questions, in the VSSQ. The main effect of prices on products shopped for had three levels, as indicated: cheap ($5–10), mid-range ($30–100), and expensive ($300–1000). Because questions 15 and 16 were included in the survey for the purpose of measuring the Cronbach's alpha variable, no analysis of variance procedure was conducted for them. Table 1 presents the summaries of results from all fourteen analyses. Table 1 indicates that of the fourteen items, only question 3.4.6, the website's funding and budget spent on security, and question 3.4.13, issues concerning the web browser, had significant differences in scores between the different price range groups (F value = 3.86, p-value = 0.034 for question 3.4.6; F value = 3.610, p-value = 0.041 for question 3.4.13). In all of the remaining items, there were no significant differences between the three groups. The results primarily indicate that the majority of e-shoppers gave just about equal importance to the majority of security items, regardless of the prices of goods they are shopping for. The findings lead to the conclusion that the e-shoppers perceive the different elements of security during shopping online to be of just about the same importance and do not make a difference in the relative importance of the security items based on the prices of goods they are shopping for. Significant differences between the treatments being present in only questions 3.4.6 and 3.4.13 begs the question regarding what is special about these items. However, on closer inspection, the statements 'The website's funding and budget spent on security' and 'Issues concerning my web browser' can be perceived as more general and imprecise explanations in comparison to other, more precise explanations of specific security items. It is therefore concluded that the significant differences among the groups for these particular items are not due to the differences between the groups in prices of goods that are being shopped for, but simply due to the general nature of the descriptions of these questions. Therefore, it can be concluded that the

Table 1. Analysis of variance results for differences for each item of the virtual shopping security questionnaire (VSSQ) between the three groups (shopping for cheap, mid-priced, and expensive products). N = 10 for each group.

Q1: Blocking of Unauthorized Access				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	5.7	2.21	1.28	0.294
Mid-Priced	6.2	1.87		
Expensive	6.9	0.32		

Q2: Emphasis on Network Security				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	5	1.89	1.56	0.229
Mid-Priced	5.5	2.01		
Expensive	6.3	0.82		

Q3: Top Management Commitment				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	6	1.25	1.15	0.330
Mid-Priced	5.6	2.46		
Expensive	6.7	0.67		

Q4: Effort to Make Users Aware of Security Procedure				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	4.8	1.48	1.58	0.225
Mid-Priced	5.5	1.78		
Expensive	5.9	0.74		

Q5: Emphasis on Login Name and Password Authentication				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	4.5	1.96	1.79	0.186
Mid-Priced	4.6	2.22		
Expensive	5.9	1.20		

Q6: Funding and Budget Spent on Security				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	4.2	1.69	3.86	0.034
Mid-Priced	4.9	2.02		
Expensive	6.2	1.03		

Q7: Monitoring of User Compliance with Security Procedures				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	5.1	1.60	0.19	0.824
Mid-Priced	5.3	1.77		
Expensive	5.5	0.71		

Q8: Distribution of Security Items within the Site				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	4.7	1.42	2.31	0.118
Mid-Priced	5.6	1.35		
Expensive	5.8	0.79		

Q9: Integration of State-of-the-Art Systems				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	5.3	1.57	1.85	0.177
Mid-Priced	5.6	1.84		
Expensive	6.5	0.71		

Q10: Web Site's Keeping Up-to-Date with Product Standards				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	5.4	1.51	0.68	0.516
Mid-Priced	5.6	1.58		
Expensive	6.1	0.99		

Q11: Web Site's Emphasis on Security in File Transfers				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	5.7	1.49	0.44	0.648
Mid-Priced	5.2	1.93		
Expensive	5.8	1.03		

Q12: Web Site's Encryption Strategy				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	5.4	1.43	1.41	0.263
Mid-Priced	5.6	1.26		
Expensive	6.3	1.06		

Q13: Issues Concerning the Web Browser				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	4.6	1.35	3.61	0.041
Mid-Priced	5.3	1.89		
Expensive	6.3	0.82		

Q14: Consolidation with Network Security Vendors				
Groups	***Mean***	***Std. Dev.***	***F-value***	***P-value***
Cheap	4.9	1.52	3.05	0.064
Mid-Priced	4.9	1.85		
Expensive	6.3	0.82		

importance of perceived security items by the users is not affected by the prices of the items being shopped for.

4.2 *Analysis of correlations among perceived security items*

Next, several correlation analyses were conducted among the perceived security items in order to understand the interrelationships between the items. Since there were no significant differences in scores between the groups with different prices for the goods, a correlation matrix between the fourteen items of perceived security using the data of all thirty participants was produced. The last two items in the questionnaire were not included in the correlation matrix again due to their inclusion being for measuring the Cronbach's alpha variable purposes. This correlation matrix is presented on table 2.

All of the correlations that are discussed in this section are significant at 0.05 confidence level. Table 2 indicates that the strongest correlation is detected between questions 3.4.2 and 3.4.4, namely, 'the website's emphasis on network security' and 'the website's effort to make its users aware of the security procedures it utilizes' ($R = 0.85$, p-value < 0.0001). The finding indicates a strong relationship between the level of emphasis a shopping site puts on security and what kind of effort the same site spends on making its users aware of these efforts. This finding is a strong indication of the perception of the existence of the security measures not a standalone item of importance for perceived security in e-shopping; making the users aware of these measures is an item of equal or close importance to shoppers that closely relates to, and possibly complements, the security emphasis of the site. In other words, the emphasis of the site on security and making the users aware of this emphasis are two highly related items for the users' overall feeling of security and are closely related to each other.

The second highest correlation was detected between questions 3.4.6 and 3.4.14, namely, 'the website's funding and budget spent on security' and 'the website's consolidation with network security vendors' ($R = 0.84$, p-value < 0.0001). The finding indicates a positive relationship between the site's security budget size and how this budget is spent. It is an indication that the e-shoppers are keen on the connection between whether the website is spending on security issues, as well as how they are spending this money, in the form of whether they are consolidating with security vendors. This strong correlation is deemed an important finding because it indicates an expectation of detail on the e-shoppers' behalf regarding the spending made for security by the website.

The third strongest correlation was found between questions 3.4.9 and 3.4.10, namely, 'the website's integration of state-of-the-art security systems' and 'the website's keeping up-to-date with product standards' ($R = 0.80$, p-

Table 2. Correlation matrix among the fourteen items of the virtual shopping security questionnaire (VSSQ) (correlations in bold are significant at 0.05 alpha level).

Perceived security items	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**	**Q11**	**Q12**	**Q13**	**Q14**
Q1: Blocking of Unauthorized Access	1.000													
Q2: Emphasis on Network Security	0.517	1.000												
Q3: Top Management Commitment	0.593	**0.657**	1.000											
Q4: Effort to Make Users Aware of Security Procedures	0.409	0.852	**0.539**	1.000										
Q5: Emphasis on Login Name and Password Authentication	0.396	0.237	0.332	0.242	1.000									
Q6: Funding and Budget Spent on Security	**0.683**	0.241	0.395	0.281	**0.631**	1.000								
Q7: Monitoring of User Compliance with Security Procedures	**0.780**	0.243	0.467	0.232	0.405	0.638	1.000							
Q8: Distribution of Security Items within the Site	0.399	0.214	0.229	0.277	0.300	0.559	0.325	1.000						
Q9: Integration of State-of-the-Art Systems	**0.741**	0.512	**0.611**	0.491	0.426	0.627	0.560	0.475	1.000					
Q10: Web Site's Keeping Up-to-Date with Product Standards	**0.688**	0.527	**0.702**	0.540	0.399	**0.661**	0.483	0.461	**0.796**	1.000				
Q11: Web Site's Emphasis on Security in File Transfers	0.398	0.323	0.367	0.244	0.376	0.312	0.147	0.194	**0.636**	0.539	1.000			
Q12: Web Site's Encryption Strategy	0.553	0.322	0.519	0.204	0.427	**0.675**	0.583	0.478	**0.661**	0.668	0.395	1.000		
Q13: Issues Concerning the Web Browser	0.601	**0.762**	0.648	0.691	0.354	0.422	0.295	0.274	**0.618**	0.630	0.345	0.503	1.000	
Q14: Consolidation with Network Security Vendors	0.623	0.226	0.440	0.287	**0.675**	**0.837**	**0.739**	0.450	**0.637**	0.520	0.290	**0.648**	0.394	1.000

value < 0.0001). This finding also indicates a strong level of demand on behalf of the users for information regarding what the vendors are doing for their security. According to the users, the integration of state-of-the-art security systems is strongly related to constantly updating these systems to the latest security standards. Again, the finding is a strong indication of the fact that the users do not only find a static security system to be present on the e-commerce site's body, but they also require to know how this security system is being used. In this case, users find a strong relationship between the site's using state-of-the-art security technology and keeping this technology up-to-date.

The fourth highest correlation has been detected between questions 3.4.1 and 3.4.7, namely, 'the website's blocking of unauthorized access into its headquarters' and 'the website's monitoring whether the users are complying with the security procedures' ($R = 0.78$, p-value < 0.0001). The finding indicates a strong relationship between those security items that mostly involve third party intrusions. It indicates a relationship between whether the website blocks unauthorized access and whether it monitors users regarding whether they are taking legitimate actions on the site quarters. The finding is also an indication of the users expecting to be given information on whether and how the site prevents possible intrusion from outside attackers.

'The website's emphasis on network security' (question 3.4.2) and 'issues concerning the user's web browser' (question 3.4.13) are also strongly correlated, having the fifth highest correlation ($R = 0.76$, p-value < 0.0001). These two items are more general in definition and the high correlation between them indicates a strong relationship between the site's emphasis on security and the web browser's capabilities in handling security in the user's (client) side, i.e. compatibility with the encryption the website requires for secure transactions.

Again, the finding indicates a relationship between the site end and user end of the security construct. Therefore, it can be argued that the users do not only expect the site's taking the security precautions for safe shopping and making them aware of those precautions, they also connect those security issues with those at the user's end, possibly, for example, to how these security precautions are compatible with the user's browser.

The correlations discussed above indicated a number of strong relationships among the individual security items including strong relationships between the site's taking the security precautions necessary and making the users aware of those precautions. Also, a high level of emphasis is given on not only what security precautions are available on the site, but also how these security items are actively engaged within the site in order to provide a site for them where the perceived security level is high, in other words where they can shop with a secure feeling. Therefore, it was concluded that clustering these security items would help understand the main groups of items that can broadly define the main categories of perceived security, eliminate security items that were conceived in the questionnaire but deemed irrelevant to perceived security by the users, as well as help develop design guidelines for future studies. This clustering method and the obtained results are described in the next section.

4.3 *Factor analysis to determine the perceived security clusters*

To identify the main clusters of items affecting perceived security in e-commerce, multivariate analysis in the form of principal components analysis was used. The principal components analysis was chosen mainly because of its relative accuracy in identifying initial factors compared to the maximum likelihood procedure (Ozok and Salvendy 2001). The fourteen items of perceived security from the questionnaire were taken as initial factors for the principal components procedure while the last two items were left out of the analysis since they were included in the questionnaire for measuring the Cronbach's alpha variable. For more information on the use of principal components analysis on survey data, see Wei and Salvendy (2000).

The factor analysis consists of two main procedures: the first procedure involves determining the number of factors to be used in the analyses, and the second part involves attributing the individual items to the factors. When more than two factors were used in the analysis, the communality in the analysis became greater than one. The analysis was repeated with up to ten factors, and the communality remained greater than ten in all cases, eliminating the possibility that the chosen number of factors could be too low. Additionally, a power analysis at the beginning had indicated ten participants per group were adequate for the experimentation. Therefore, the reason of the communality being greater than one could also not be the low number of participants. Five additional analyses were conducted for determining the optimal number of factors for the analysis: scree plot, Schwarz's Bayesian criterion, Akaike's information criterion, Bartlett's chi-square criterion and comprehensibility criterion (Ozok and Salvendy 2001). With the exception of scree plot and comprehensibility criterion, none of these criteria gave a meaningful output for the optimal number of factors. The scree plot showed a sharp bend starting at two factors. The comprehensibility criterion showed two factors to be the best number for easily comprehensible meanings. Therefore, two factors were determined to be the optimal number for the analysis.

Prior to the second procedure, an orthogonal rotation of the factors called Varimax rotation was conducted on the initial fourteen variables, mainly because the Varimax rotation is used for making the factors more suitable for clustering. Similar to the work by Wei and Salvendy (2000)

and Ozok and Salvendy (2001), a threshold of 0.40 loading was used to conclude when a particular security item belonged to a certain factor. The factor loadings of the fourteen items on the two factors are presented on table 3.

Next, the variance caused by the two factors was inspected. The percentages of the overall variance caused by the individual factors are fairly close, with 59.1% being caused by factor 1 and 40.9% being caused by factor 2. It should be noted that this variance is the part of the overall variance coming from factors only, and the percentages therefore add up to one. Table 3 indicates that in fourteen security items (which can also be called variables), three items had a weight higher than 0.40 on both factors instead of one factor. These items were question 3.4.1 (0.68 for factor 1, 0.50 for factor 2), question 3.4.9 (0.63 for factor 1 and 0.60 for factor 2), and question 3.4.10 (0.55 for factor 1 and 0.66 for factor 2). In these cases, the particular item was attributed to the factor on which it has a higher weight than the other, similarly to the study by Ozok and Salvendy (2001). Additionally, since only 20% (three out of fourteen) of the items had such weight distribution with a weight higher than 0.40 on two factors, it was deemed appropriate to distribute those items to the factor they had the higher weight on.

After the clustering process, eight items were grouped into factor 1 and six items were grouped into factor 2. The items that were grouped into factor 1 included: 'the website's blocking of unauthorized access', 'the website's emphasis on login name and password request for secure areas', 'the website's funding and budget spent on security', 'the website's monitoring whether the users are complying with the security policies', 'the website's distribution of security items within its body', 'the website's integration of state-of-the-art security systems', 'the website's encryption strategy', and 'the website's consolidation with network security vendors'. The items that were grouped into factor 2 included: 'the website's emphasis on network security', 'the website's top management commitment in security-related issues', 'the website's effort to make its users aware of the security procedures it utilizes', 'the website's keeping up-to-date with product standards', 'the website's emphasis on security in file transfers over the internet', and 'issues concerning the web browser'.

Close inspection of the items grouped into factor 1 indicates a common aspect among these security items, namely, they are all parts of the actions a website can take to ensure that the users feel secure during the time they are shopping on its quarters. It can be concluded that the expectations concerning the operational actions the website should take in order for the online shoppers to have a relatively high level of perceived security include the site's blocking of unauthorized access, its emphasis on login name and password requests, monitoring whether users are complying with the security policies, its distribution of security items within its body, its integration of state-of-the-art security technologies, and its encryption strategy. Although the website's funding and budget spent on security cannot be thought of as an item directly concerning its actions, it should be noted that the budget item is concerned with the core implementation of the operations concerning these perceived security items. Similarly, the website's consolidation with network security vendors no doubt is an action directly towards improving security operations on their site to provide the latest security technologies on their site. For these reasons, factor 1 has been named the 'perceived operational factor' concerning perceived security in e-shopping.

While factor 1 is concerned with operational items, the elements grouped into factor 2 show a pattern of items that are concerned with more broader-level, policy-related security items. These items included 'the website's emphasis on network security', 'the website's top management commitment in security-related issues', 'the website's effort to make its users aware of the security procedures it utilizes', 'the website's keeping up-to-date with product

Table 3. Factor loadings for the items of the virtual shopping security questionnaire (VSSQ).

Perceived security items	Factor1	Factor2
Q1: Blocking of Unauthorized Access	**0.68**	0.50
Q2: Emphasis on Network Security	0.07	**0.92**
Q3: Top Management Commitment	0.38	**0.73**
Q4: Effort to Make Users Aware of Security Procedures	0.07	**0.86**
Q5: Emphasis on Login Name and Password Authentication	**0.67**	0.16
Q6: Funding and Budget Spent on Security	**0.88**	0.19
Q7: Monitoring of User Compliance with Security Procedures	**0.78**	0.16
Q8: Distribution of Security Items within the Site	**0.59**	0.17
Q9: Integration of State-of-the-Art Systems	**0.63**	0.60
Q10: Web Site's Keeping Up-to-Date with Product Standards	0.55	**0.66**
Q11: Web Site's Emphasis on Security in File Transfers	0.30	**0.45**
Q12: Web Site's Encryption Strategy	**0.73**	0.31
Q13: Issues Concerning the Web Browser	0.27	**0.83**
Q14: Consolidation with Network Security Vendors	**0.90**	0.15

standards', and 'the website's emphasis on security in file transfers over the internet'. The only item that does not have a direct connection with the website's policy-related factors seems to be the last item in the group, namely, 'issues concerning the web browser'. However, this inclusion can be explained by the possibility that users may have interpreted this general in nature question as how the website addresses security issues that may be stemming from the user's website, such as bits of encryption that is a common issue where users are denied access to a secure site when their browser does not have the required encryption (today's standard for minimum is 128 bits). For these reasons, factor 2 has been named the 'perceived policy-related factor' concerning perceived security in e-shopping.

It is believed by the researchers that the induction of the two types of factors affecting perceived security of e-commerce will allow for the development of a set of guidelines concerning perceived security in e-commerce. In other words, the factors developed can work as categories of perceived security in e-commerce. The results of the multivariate analysis indicates that practically, these categories identify the main issues that e-commerce can concentrate on to provide perceived security for their customers, thereby improving the overall shopping experience and increasing sales. With the lowest mean of scores for a particular question being 5.0 out of 7 (the website's emphasis on login name and password request for entering secure areas), it can be concluded that there is not an item among those covered by the questionnaire that was not of significant importance according to the participants. Therefore, table 4 presents the items of importance to provide perceived security to customers in online shopping.

5. Generating guidelines for perceived security in e-commerce among Turkish student shoppers

The different analyses indicated that all of the conceptually identified elements of perceived security were significant from the users' perspective in terms of their role in providing a secure feeling in the shopping environment for the e-shoppers. Therefore, using the items identified within the virtual shopping security questionnaire and results from the statistical analyses, fourteen guidelines for e-commerce sites in providing perceived security for their shoppers have been identified in two categories by translating the VSSQ items into explicit guidelines.

The following guidelines do not indicate the actual security precautions that are suggested to be taken by e-commerce sites, but rather the security items that are to be presented to the user to provide perceived security during Gshopping.

Guidelines

Inform the user shopping on your site that:

1. **Any possible unauthorized access into the site quarters is blocked (operational).**

The current survey as well as previous studies (Miyazaki and Fernandez 2000) indicated that one of the major items affecting perceived security in e-commerce is the awareness by the users regarding the possibility of theft of sensitive information by third-party intrusion. This guideline therefore suggests that the user should be made aware of the fact that such intrusion has been made impossible. From the practical point of view, this can be done by presenting a message that can say: 'You are entering a secure area. All unauthorized access to this area is blocked.'

2. **The website puts significant emphasis on security (policy-related).**
3. **The site's top management is entirely committed to security items (policy-related).**
4. **A significant effort is spent by the website to apply the security procedures (policy-related).**

Since the results of the current study indicated that it is important for customers to know that security is an issue being taken seriously by the website at a management level, these guidelines indicate that customers should be made aware of the emphasis and effort given to security at a top-management level by the company. This can be presented as a policy statement by the company at the entrance to the site with a message saying 'This company is entirely

Table 4. The newly created factors and their corresponding perceived security items obtained from the factor analysis.

Perceived operational factors	Perceived policy-related factors
Blocking of Unauthorized Access	Emphasis on Network Security
Emphasis on Login Name and Password Authentication	Top Management Commitment
Funding and Budget Spent on Security	Effort to Make Users Aware of Security Procedures
Monitoring of User Compliance with Security Procedures	Web Site's Keeping Up-to-Date with Product Standards
Integration of State-of-the-Art Systems	Web Site's Emphasis on Security in File Transfers
Distribution of Security Items within the Site	Issues Concerning the Web Browser
Web Site's Encryption Strategy	
Consolidation with Network Security Vendors	

committed at a top management level to providing full security for its customers by applying security procedures.'

5. **Login and password authentications are used extensively to provide secure transaction environments (operational).**

The importance of password-based authentication is a non-trivial issue. For this reason, this particular guideline indicates a high level of login and password-based authentication use within the website body. Making the user aware of the authentication system can obviously be accomplished by the sole availability of these systems.

6. **The website has significant budget and funding spent on security (operational).**
7. **Users are monitored for complying with the site's security policies (operational).**

Guideline strongly relates to guideline 1. Warnings concerning monitoring unauthorized access can be presented to the user by putting up warnings regarding the site's monitoring activities. However, a 'big brother' feeling should be avoided by not overdoing this conveying of information, possibly by only giving one warning sign that users are monitored for complying with the website's security procedures.

8. **Security items are distributed consistently within the site (operational).**

A consistent distribution of security items within the shopping site ensures the user to have the same high amount of perceived security at different parts of the site shopping for different products. For this purpose, this particular guideline suggests that the same amount of security items such as secure payment methods and warnings like those suggested in the previous guidelines should be presented throughout the entire site.

9. **The website has integrated state-of-the-art security systems (operational).**
10. **The website is always keeping up-to-date with the product standards (policy-related).**

Since the website's keeping up-to-date with state of the art technology for security has been proven to be an item affecting perceived security, guideline suggests making the user aware of that fact. Again, it is suggested that this conveying of information can be done with a short message regarding the system, for example what version of the security software is being used.

11. **The website thoroughly consolidates with security vendors (operational).**
12. **All file transfers are made over secure internet communication lines (policy-related).**
13. **The site has a strict encryption strategy in online transactions (operational).**

Similar to the previous items, as part of overall security, the information concerning secure file transfers can be conveyed by a message at the beginning of file transfers and payment transactions ensuring the shoppers that files and information are transferred securely and in encrypted format.

14. **The user's web browser information is checked by the website (policy-related).**

The website's retrieving some information regarding the user can ensure users that they are taking the appropriate precautions for security. Therefore, presenting their own browser information to the users is believed to be a perceived security-boosting factor.

6. Conclusion, recommendations and future directions

The concept of security has been the concern of consumers, e-commerce companies and researchers alike. What is presented on the screen concerning security to the user during shopping is concluded by this study to significantly affect how secure users feel while shopping online. This study identified two main categories of perceived security: perceived operational, and perceived policy-related factors. The perceived operational factors are hands-on practical actions the users expect the companies to employ to ensure their security. The perceived policy-related factors are rules and principles adopted and enforced by companies to prevent any harm to site and user security, which are two directly related items. It is concluded that the perceived security can be boosted in online shopping sites if the guidelines for perceived security items are followed by shopping sites.

It is also concluded that the college student users consider strong relationships among the security items and the site's effort to make its users aware of these security items. A company-level dedication to security is expected and is integral to perceived security, with the company's emphasis on security strongly correlating to making the users aware of the procedures the site is using, the size of the company's security budget strongly correlating to its consolidation with vendors, and the integration of state-of-the-art security systems strongly correlating to keeping up-to-date with the product standards. On the operational side, the users indicated a strong correlation between blocking of unauthorized access and monitoring security. All these items indicate a high level of demand by the customers to have an understanding of what security measures there are while they are shopping on e-commerce sites. One additional finding was that the users expected the

same level of security from the sites, regardless of the prices of products they are shopping for.

The validity of the findings should be handled with caution because of the specific scope of the participants. Due to some practical difficulties discussed before, this study concentrated on perceived security issues concerning mostly male Turkish college student shoppers. Therefore, the group of participants was limited to a segment of the overall Turkish e-shopper population. However, it should be noted that the validity of every study is arguably limited to the participant group the study was conducted on. While the researchers have no evidence (based on the literature search and the findings of this study) on whether there are significant differences in perceived security among male and female engineering college students in Turkey, the only population group the findings of this study apply to with empirical support is Turkish, male engineering college students. The participant group was to some extent consistent with the participant group used in the study by Lightner *et al*. (2002). That particular study also used mostly male (64%) Turkish engineering college students. However, the low number of female participants prevented the researchers in the current study from conducting any statistical analyses contrasting the two genders. A more gender-balanced study can be done in the future to determine whether significant differences between male and female populations concerning perceived security exist.

It should be noted that there are a variety of items that can be considered as contributing to the improvement of security in online shopping. Offering alternative payment and shipping services, giving the shoppers options to opt-in or opt-out of mailing lists, security certifications such as Verisign ensuring trustworthiness of the site, name recognition, or the site's history on whether it has been hacked into in the past, are all items that can be viewed as influencing the feeling of security of online shoppers. However, these items have not been covered in this research, but rather items presented to the users during shopping on the interface are concentrated on. Future research may include these option-related and certification items and comprehensively explore perceived security from all perspectives beyond presentations on the interface. Additionally, it should be noted that the information was collected after users had completed their shopping in a simulated environment. The researchers believe that since the users are already avid e-shoppers, the virtual shopping experience familiarized them with a common shopping environment for a certain product price range, and the results effectively reflected their real life opinions. However, future research can test the implementation of the developed guidelines by presenting the questionnaire items to actual shoppers, possibly on e-commerce sites after they have completed an actual shopping transaction. Finally, the results of this study can be compared to those of a similar population from a different culture.

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